Diagonalization

Geometric Algorithms

CAS CS 132

Objectives

» Finish our discussion on the characteristic polynomial

» Motivate diagonalization via linear dynamical systems and changes of coordinate systems

» Describe how to diagonalize a matrix

Keywords

multiplicity

similar matrices

diagonalizable matrices

change of basis

eigenbasis

Recap: Characteristic Polynomial

Recall: Determinants and Invertibility

Recall: Determinants and Invertibility

$\det(A)$ is an value associate with the matrix A

Recall: Determinants and Invertibility

$\det(A)$ is an value associate with the matrix A

Thm. A matrix is invertible if and only if $\det(A) \neq 0$

Recall: Determinants and Invertibility

$\det(A)$ is an value associate with the matrix A

Thm. A matrix is invertible if and only if $\det(A) \neq 0$

So by the Invertible Matrix Theorem:

Recall: Determinants and Invertibility

$\det(A)$ is an value associate with the matrix A

Thm. A matrix is invertible if and only if
$\det(A) \neq 0$

So by the Invertible Matrix Theorem:

$$\det(A - \lambda I) = 0 \quad \equiv \quad (A - \lambda I)\mathbf{x} = \mathbf{0} \text{ has nontrivial solutions}$$

$$\equiv \quad \lambda \text{ is an eigenvalue of } A$$

Recall: Determinants and Invertibility

$\det(A)$ is an value associate with the matrix A

Thm. A matrix is invertible if and only if $\det(A) \neq 0$

So by the Invertible Matrix Theorem:

$\det(A - \lambda I) = 0$ $\equiv$ $(A - \lambda I)\mathbf{x} = \mathbf{0}$ **has nontrivial solutions**

polynomial in λ

$\equiv$ λ is an eigenvalue of A

How To: Finding Eigenvalues

How To: Finding Eigenvalues

Q. Determine the eigenvalues of A

How To: Finding Eigenvalues

Q. Determine the eigenvalues of A

A. Find *roots* of the *characteristic poly* of A:

How To: Finding Eigenvalues

Q. Determine the eigenvalues of A

A. Find *roots* of the *characteristic poly* of A:

$$\det(A - \lambda I)$$

Example

$$A = \begin{bmatrix} 1 & -1 \\ \cancel{1} & -3 \end{bmatrix}$$

4

$$A - \lambda I = \begin{bmatrix} 1-\lambda & -1 \\ 4 & -3-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (1-\lambda)(-3-\lambda) - (-4)$$

$$= (\lambda - 1)(\lambda + 3) + 4$$

$$= \lambda^2 + 2\lambda - 3 + 4$$

$$= \lambda^2 + 2\lambda + 1 = (\lambda + 1)^2 \quad \boxed{\lambda = -1}$$

An Observation: Multiplicity

$$\boxed{\lambda^1}(\lambda - 1)^{\boxed{2}}(\lambda - 4)^{\boxed{1}}$$ multiplicities

An Observation: Multiplicity

$$\boxed{\lambda^1}(\lambda - 1)^{\boxed{2}}(\lambda - 4)^{\boxed{1}}$$ multiplicities

In the examples so far, we've seen a number appear as a root multiple times

An Observation: Multiplicity

$$\boxed{\lambda^1}(\lambda - 1)^{\boxed{2}}(\lambda - 4)^{\boxed{1}}$$ multiplicities

In the examples so far, we've seen a number appear as a root multiple times

This is called the **algebraic multiplicity** of the root

An Observation: Multiplicity

$$\boxed{\lambda^{1}}(\lambda - 1)^{\boxed{2}}(\lambda - 4)^{\boxed{1}}$$ multiplicities

In the examples so far, we've seen a number appear as a root multiple times

This is called the **algebraic multiplicity** of the root

Is the algebraic multiplicity meaningful?

Multiplicity and Dimension

Thm. The dimension of the eigenspace of A for the eigenvalue λ is *at most* the algebraic multiplicity of λ in $\det(A - \lambda I)$

The algebraic multiplicity is an upper bound on "how large" the eigenspace is

Practice Problem

$$\begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix}$$

Determine the eigenvalues and an eigenbasis for the above matrix

eigenvalues of A

basis of $\mathbb{R}^2$

Answer

$$A = \begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix}$$

$$\boxed{\begin{aligned} &\lambda = 1, 6 \\ &\mathbb{R}^2 = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 4 \end{bmatrix} \right\} \end{aligned}}$$

$$\det(A - \lambda I) = \det \begin{bmatrix} (5-\lambda) & 1 \\ 4 & (2-\lambda) \end{bmatrix} = (5-\lambda)(2-\lambda) - 4$$

$$= \lambda^2 - 7\lambda + 10 - 4$$

$$= \lambda^2 - 7\lambda + 6$$

$$= (\lambda - 6)(\lambda - 1)$$

$$= \lambda = 1, 6$$

$$A - I = \begin{bmatrix} 4 & 1 \\ 4 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1/4 \\ 0 & 0 \end{bmatrix}$$

$x_1 = -\frac{1}{4}x_2$

x_2 is ~~free~~

$$\begin{bmatrix} 6 & 1 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 4 \end{bmatrix} = \begin{bmatrix} -5 + 4 \\ -4 + 8 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

$$\vec{r} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

Answer

$$\begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix}$$

$$A - 6I = \begin{bmatrix} -1 & 1 \\ 4 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

$x_1 = x_2$

x_2 is free

$$\vec{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5+1 \\ 4+2 \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \end{bmatrix} = 6 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Motivating Diagonalization via Linear Dynamical Systems

Recall: Eigenbasis

Recall: Eigenbasis

Def. An eigenbasis of H for $A \in \mathbb{R}^{n \times n}$ is a basis of H made up of eigenvectors of A

Recall: Eigenbasis

Def. An eigenbasis of H for $A \in \mathbb{R}^{n \times n}$ is a basis of H made up of eigenvectors of A

We will be almost exclusively interested of eigenbases of $\mathbb{R}^n$ when $A \in \mathbb{R}^{n \times n}$

Recall: Eigenbasis

Def. An eigenbasis of H for $A \in \mathbb{R}^{n \times n}$ is a basis of H made up of eigenvectors of A

We will be almost exclusively interested of eigenbases of $\mathbb{R}^n$ when $A \in \mathbb{R}^{n \times n}$

When can we describe any vector in $\mathbb{R}^n$ as a unique linear combination of eigenvectors of A?

Recall: Linear Dynamical Systems

$$
\begin{aligned}
\mathbf{v}_1 &= A\mathbf{v}_0 \\
\mathbf{v}_2 &= A\mathbf{v}_1 = A^2\mathbf{v}_0 \\
\mathbf{v}_3 &= A\mathbf{v}_2 = A^3\mathbf{v}_0 \\
\mathbf{v}_4 &= A\mathbf{v}_3 = A^4\mathbf{v}_0 \\
&\vdots
\end{aligned}
$$

A **linear dynamical system** describes a sequence of **state vectors** starting at $\mathbf{v}_0$

Recall: Linear Dynamical Systems

$$\mathbf{v}_1 = A\mathbf{v}_0$$
$$\mathbf{v}_2 = A\mathbf{v}_1 = A^2\mathbf{v}_0$$
$$\mathbf{v}_3 = A\mathbf{v}_2 = A^3\mathbf{v}_0$$
$$\mathbf{v}_4 = A\mathbf{v}_3 = A^4\mathbf{v}_0$$
$$\vdots$$

multiplying by A changes the state

A **linear dynamical system** describes a sequence of **state vectors** starting at $\mathbf{v}_0$

demo

Eigenbases and Closed-Form solutions

Eigenbases and Closed-Form solutions

Given $\mathbf{v}_k = A\mathbf{v}_{k-1} = A^k\mathbf{v}_0$, if

$$\mathbf{v}_0 = \alpha_1\mathbf{b}_1 + \alpha_2\mathbf{b}_2 + \alpha_3\mathbf{b}_3$$

Eigenbases and Closed-Form solutions

Given $\mathbf{v}_k = A\mathbf{v}_{k-1} = A^k\mathbf{v}_0$, if

eigenvectors of A

$$\mathbf{v}_0 = \alpha_1\mathbf{b}_1 + \alpha_2\mathbf{b}_2 + \alpha_3\mathbf{b}_3$$

Eigenbases and Closed-Form solutions

Given $\mathbf{v}_k = A\mathbf{v}_{k-1} = A^k\mathbf{v}_0$, if

$$\mathbf{v}_0 = \alpha_1 \underbrace{\mathbf{b}_1}_{} + \alpha_2 \mathbf{b}_2 + \alpha_3 \mathbf{b}_3$$

eigenvectors of A

then

$$A^k\mathbf{v}_0 = \alpha_1\lambda_1^k\mathbf{b}_1 + \alpha_2\lambda_2^k\mathbf{b}_2 + \alpha_3\lambda_3^k\mathbf{b}_3$$

Eigenbases and Closed-Form solutions

Given $\mathbf{v}_k = A\mathbf{v}_{k-1} = A^k\mathbf{v}_0$, if

$$\mathbf{v}_0 = \alpha_1 \underbrace{\mathbf{b}_1}_{} + \alpha_2 \mathbf{b}_2 + \alpha_3 \mathbf{b}_3$$

eigenvectors of A

then

eigenvalues of A

$$A^k\mathbf{v}_0 = \alpha_1 \lambda_1^k \mathbf{b}_1 + \alpha_2 \lambda_2^k \mathbf{b}_2 + \alpha_3 \lambda_3^k \mathbf{b}_3$$

Eigenbases and Closed-Form solutions

$A \in \mathbb{R}^{3\times 3}$

Given $\mathbf{v}_k = A\mathbf{v}_{k-1} = A^k\mathbf{v}_0$, if

$\lambda = 0, 1, 3$

eigenvectors of A

$$\mathbf{v}_0 = \alpha_1\mathbf{b}_1 + \alpha_2\mathbf{b}_2 + \alpha_3\mathbf{b}_3$$

$\vec{b}_1, \vec{b}_2, \vec{b}_3$

$A^k\vec{v}_0 = A^k(\alpha_1\vec{b}_1 + \alpha_2\vec{b}_2 + \alpha_3\vec{b}_3)$

then

$= \alpha_1 A^k\vec{b}_1 + \alpha_2 A^k\vec{b}_2$

eigenvalues of A

$$A^k\mathbf{v}_0 = \alpha_1\lambda_1^k\mathbf{b}_1 + \alpha_2\lambda_2^k\mathbf{b}_2 + \alpha_3\lambda_3^k\mathbf{b}_3$$

$+ \alpha_3 A^k\vec{b}_3$

closed-form solution

$\alpha_2\vec{b}_2 + 3^k\vec{b}_3$

$= \alpha_1 0^k\vec{b}_1 + \alpha_2 1^k\vec{b}_2 + \alpha_3 3^k\vec{b}_3$

Application: Eigenbases and Limiting Behavior

Thm. If A has an eigenbasis with eigenvalues

$$\lambda_1 \geq \lambda_2 \ldots \geq \lambda_k$$

then $\mathbf{v}_k \sim \lambda_1^k \mathbf{u}$ for some vector $\mathbf{u}$

The system grows exponentially in λ_1

Application: Eigenbases and Limiting Behavior

Thm. If A has an eigenbasis with eigenvalues

$$\lambda_1 \geq \lambda_2 \ldots \geq \lambda_k$$

then $\mathbf{v}_k \sim \lambda_1^k \mathbf{u}$ for some vector $\mathbf{u}$

The system grows exponentially in λ_1

The Takeaway

The Takeaway

Given a basis $\mathscr{B}$ for $\mathbb{R}^n$, we only need to know how $A \in \mathbb{R}^n$ behaves on $\mathscr{B}$

The Takeaway

Given a basis $\mathscr{B}$ for $\mathbb{R}^n$, we only need to know how $A \in \mathbb{R}^n$ behaves on $\mathscr{B}$

Sometimes, A behaves simply on $\mathscr{B}$, as in the case of *eigenbases*

The Takeaway

Given a basis $\mathscr{B}$ for $\mathbb{R}^n$, we only need to know how $A \in \mathbb{R}^n$ behaves on $\mathscr{B}$

Sometimes, A behaves simply on $\mathscr{B}$, as in the case of *eigenbases*

What we're really doing is changing our coordinate system to expose a behavior of A

Recap: Change of Basis

Recall: Bases define Coordinate Systems

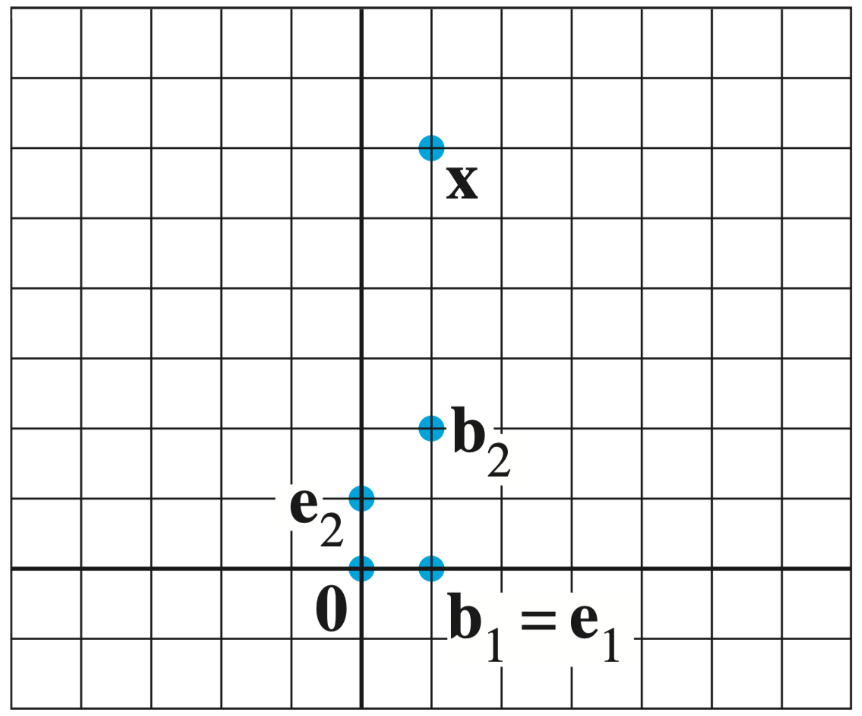


FIGURE 1 Standard graph paper.

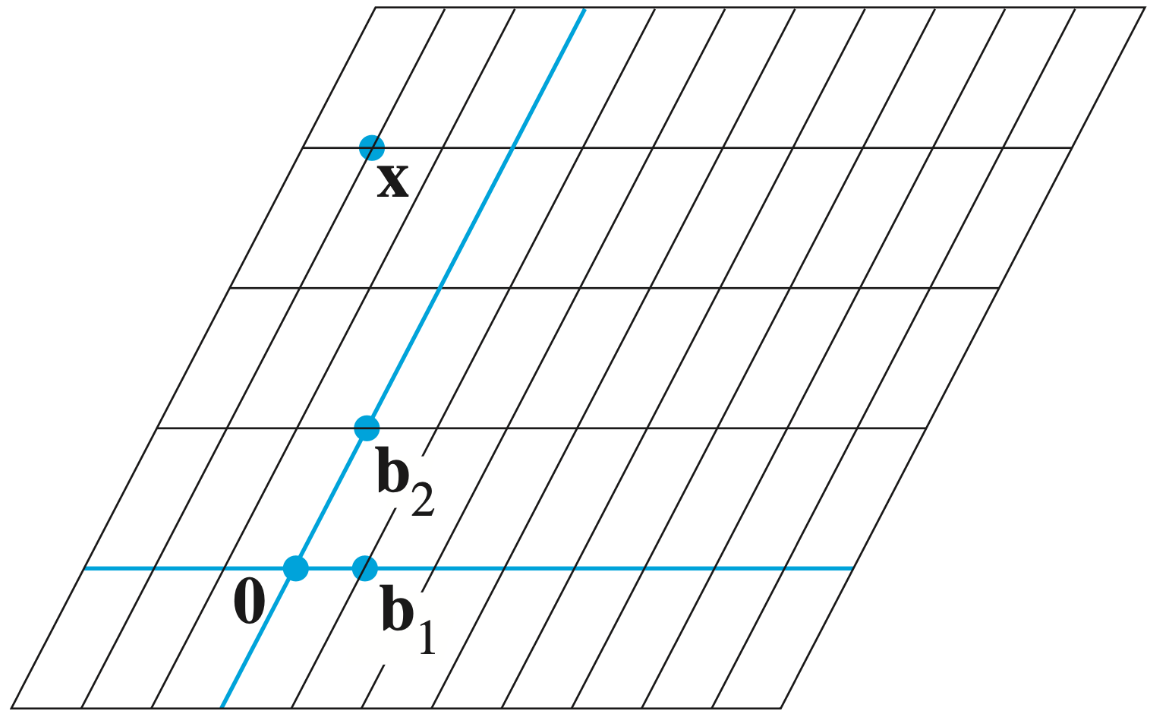


FIGURE 2 $\mathcal{B}$-graph paper.

Recall: Bases define Coordinate Systems

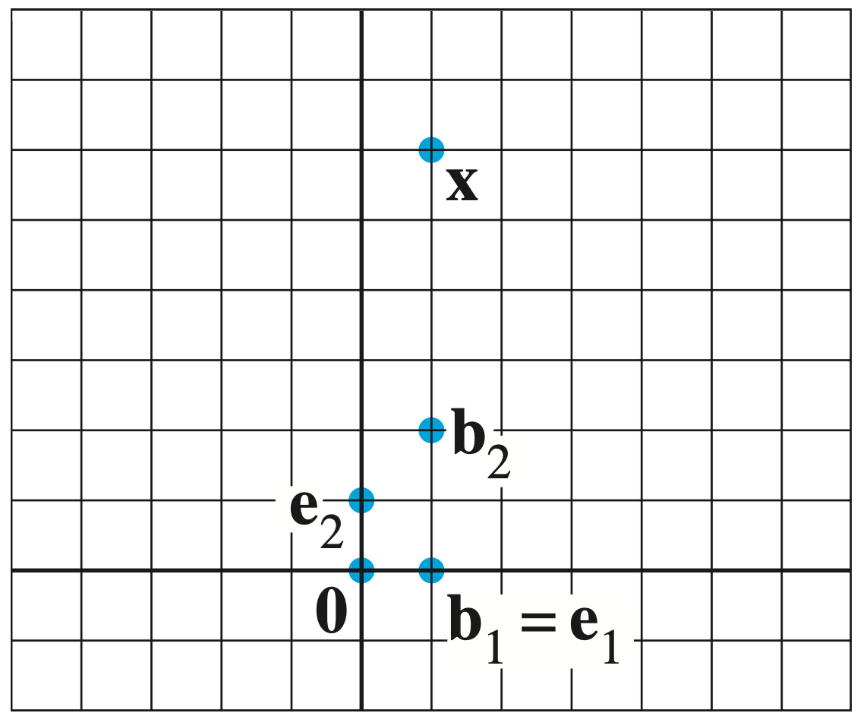


FIGURE 1 Standard graph paper.

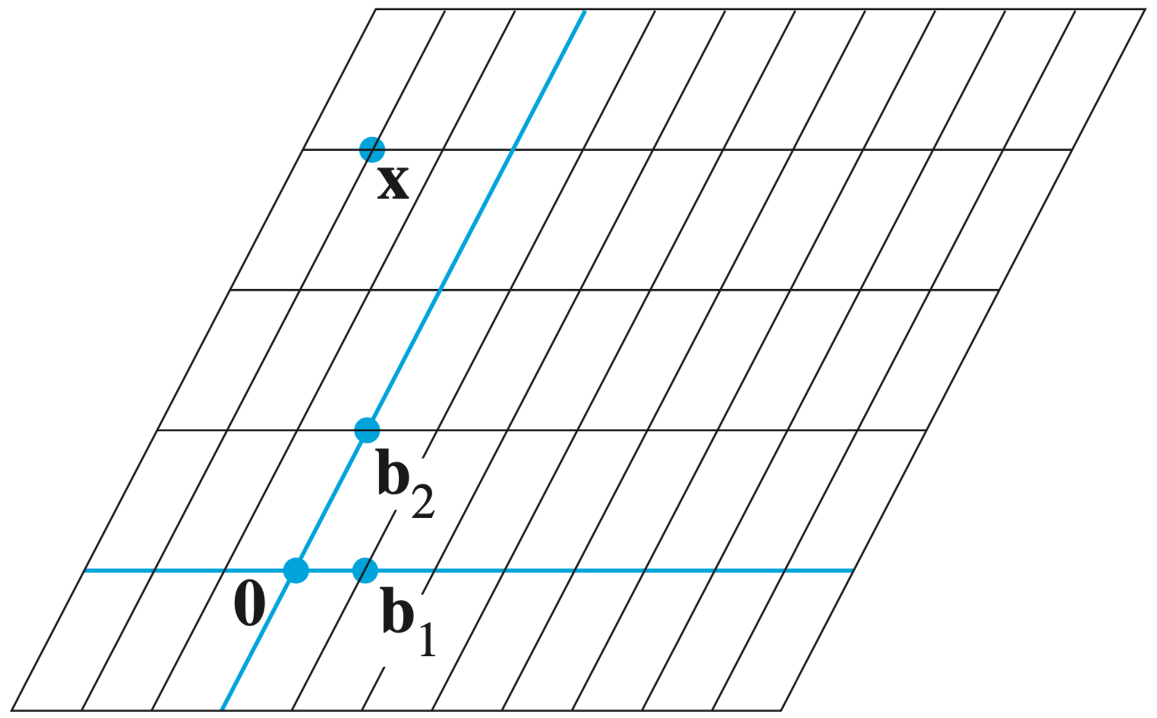


FIGURE 2 $\mathcal{B}$-graph paper.

Given a basis $\mathscr{B}$ of $\mathbb{R}^n$, there is **exactly one way** to write every vector as a linear combination of vectors in $\mathscr{B}$

Recall: Bases define Coordinate Systems

$\vec{x} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$

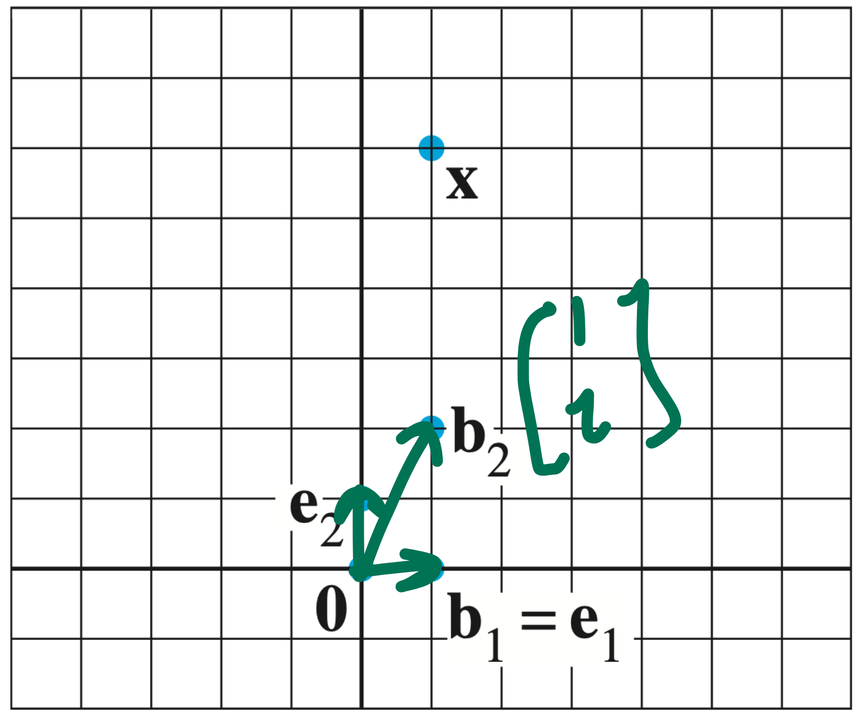


FIGURE 1 Standard graph paper.

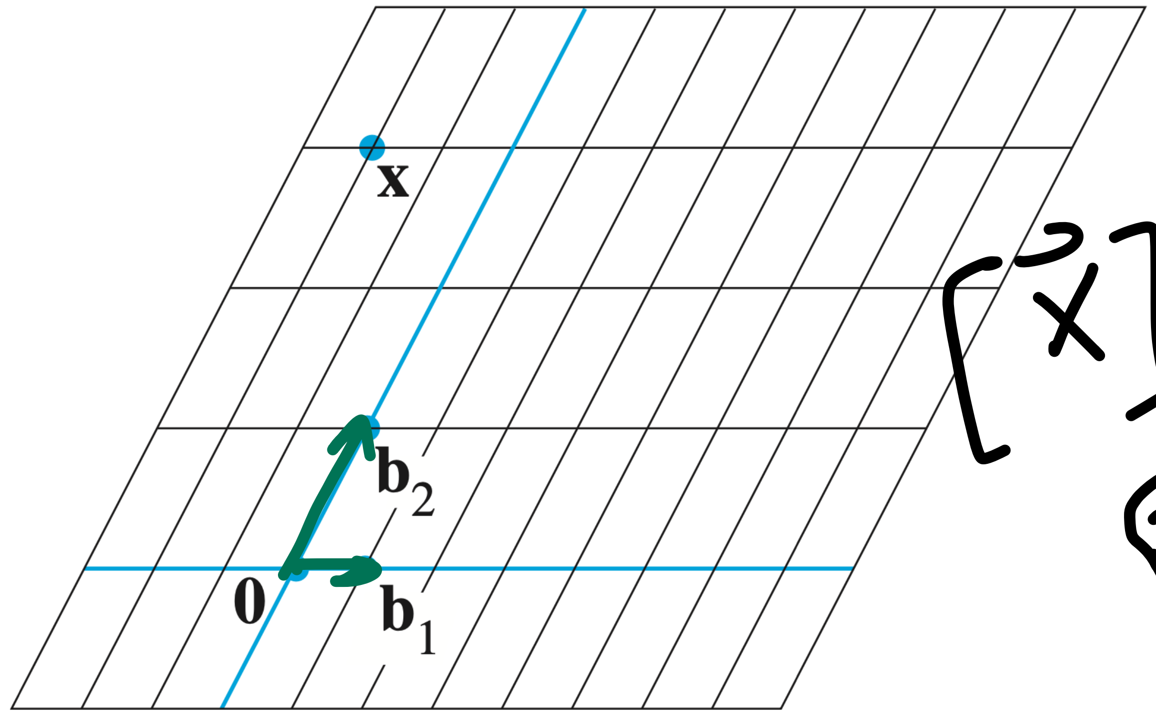


FIGURE 2 $\mathcal{B}$-graph paper.

$[\vec{x}]_{\mathcal{B}} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$

$\vec{x} = -2\vec{b}_1 + 3\vec{b}_2$

Given a basis $\mathscr{B}$ of $\mathbb{R}^n$, there is **exactly one way** to write every vector as a linear combination of vectors in $\mathscr{B}$

Every basis provides a way to write down *coordinates* of a vector

Recall: Bases define Coordinate Systems

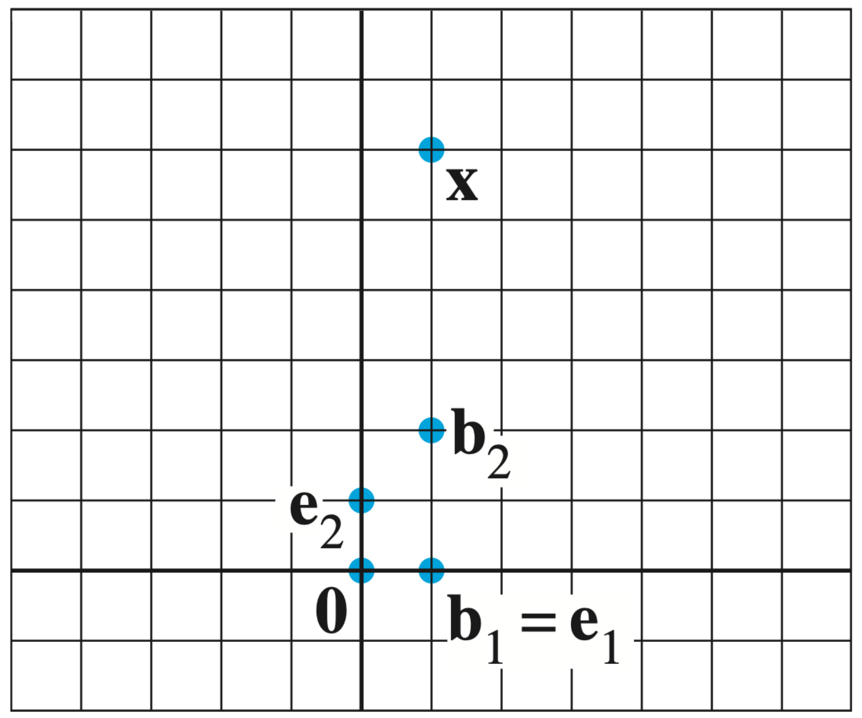


FIGURE 1 Standard graph paper.

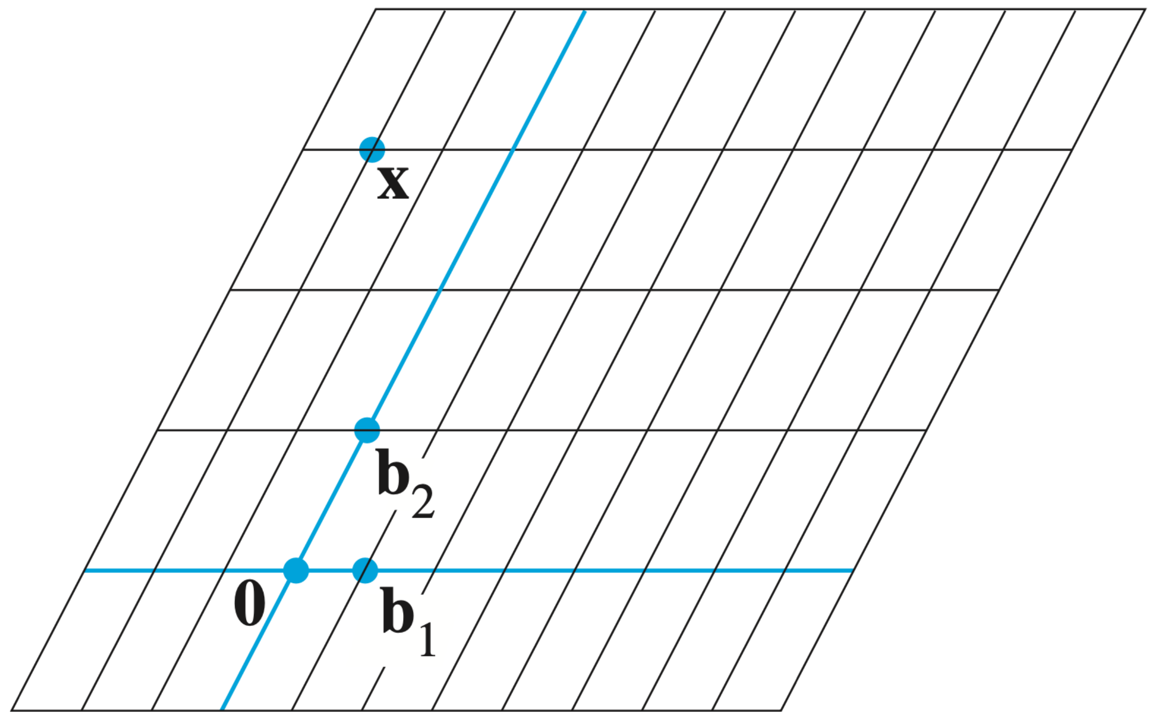


FIGURE 2 $\mathcal{B}$-graph paper.

Given a basis $\mathscr{B}$ of $\mathbb{R}^n$, there is **exactly one way** to write every vector as a linear combination of vectors in $\mathscr{B}$

Every basis provides a way to write down *coordinates* of a vector

$\mathscr{B}$ defines a "different grid for our graph paper"

Linear Algebra and its Applications (Lay, Lay, McDonald)

Recall: Coordinate Vectors

Recall: Coordinate Vectors

Let $\mathbf{v}$ be a vector in a $\mathbb{R}^n$ and let $\mathscr{B} = (\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n)$ be a basis of $\mathbb{R}^n$ where $\mathbf{v} = a_1\mathbf{b}_1 + a_2\mathbf{b}_2 + \ldots + a_n\mathbf{b}_n$

Df. The **coordinate vector of $\mathbf{v}$ relative to** $\mathscr{B}$ is

Recall: Coordinate Vectors

Let $\mathbf{v}$ be a vector in a $\mathbb{R}^n$ and let $\mathscr{B} = (\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n)$ be a basis of $\mathbb{R}^n$ where $\mathbf{v} = a_1\mathbf{b}_1 + a_2\mathbf{b}_2 + \ldots + a_n\mathbf{b}_n$

Df. The **coordinate vector of $\mathbf{v}$ relative to $\mathscr{B}$** is

$$[\mathbf{v}]_{\mathscr{B}} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

$\vec{v} \mapsto [\vec{v}]_\beta$

is <u>linear</u>

Question

We know that if a $n \times n$ matrix $B = [\mathbf{b}_1 \ \mathbf{b}_2 \ \ldots \ \ \mathbf{b}_n] \in \mathbb{R}^{n \times n}$ is invertible, then the columns of B form a basis $\mathscr{B}$ of $\mathbb{R}^n$

What is the matrix that implements the transformation

$\vec{e}_1 \mapsto [\vec{e}_1]_B$

$\vec{e}_2 \mapsto [\vec{e}_2]_B$

$\vdots$

$$\mathbf{x} \mapsto [\mathbf{x}]_{\mathscr{B}} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

$T(\vec{e}_1) = \vec{b}_1$

$1\vec{b}_1 + 0 \cdot \vec{b}_2 + \ldots 0 \vec{b}_n$

$T(\vec{e}_2) = \vec{b}_2$

where $\mathbf{x} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \ldots + c_n\mathbf{b}_n$?

Change of Basis Matrix

Thm. If $\mathscr{B} = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n\}$ form a basis of $\mathbb{R}^n$, then

$$[\mathbf{x}]_{\mathscr{B}} = [\mathbf{b}_1\ \mathbf{b}_2\ \ \ldots\ \ \mathbf{b}_n]^{-1}\mathbf{x}$$

Matrix inverses perform changes of bases

How To: Change of Basis

Q. Given a basis $\mathscr{B} = (\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n)$ of $\mathbb{R}^n$, find the matrix which implements $\mathbf{x} \mapsto [\mathbf{x}]_{\mathscr{B}}$

A. Construct the matrix $[\mathbf{b}_1\ \mathbf{b}_2\ \ldots\ \mathbf{b}_n]^{-1}$

Example

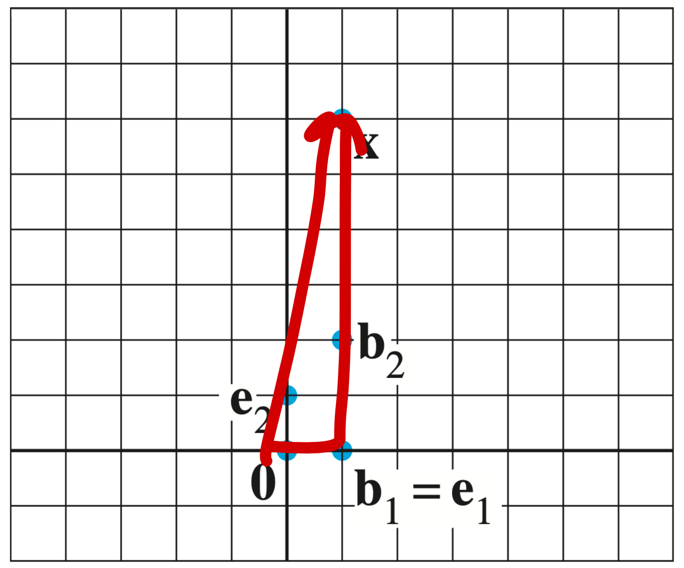


FIGURE 1 Standard graph paper.

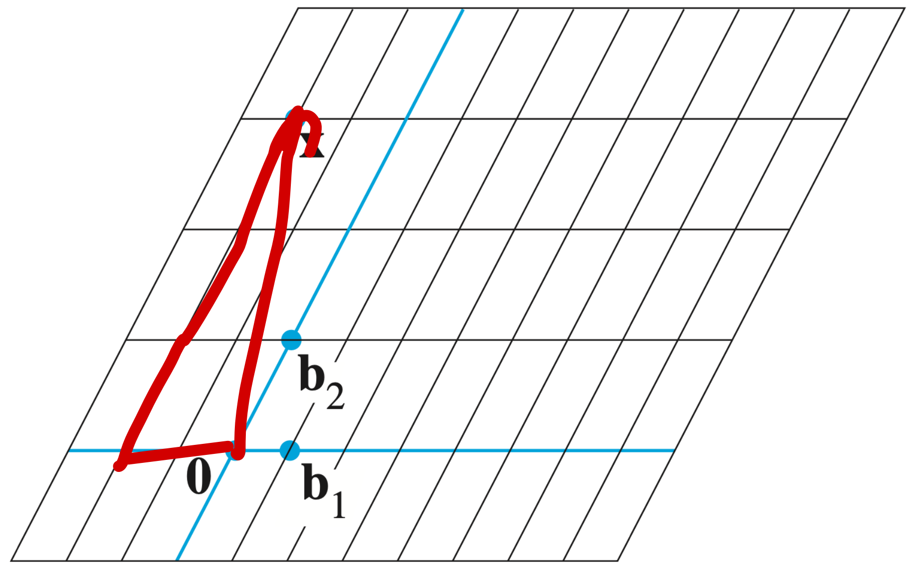


FIGURE 2 $\mathcal{B}$-graph paper.

Write the change-of-bases matrix for the basis $\mathcal{B} = \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right)$

$$\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}^{-1} = \frac{1}{2-0}\begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1/2 \\ 0 & 1/2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1/2 \\ 0 & 1/2 \end{bmatrix}\begin{bmatrix} 1 \\ 6 \end{bmatrix} = \begin{bmatrix} 1-3 \\ 0+3 \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

Diagonalization

Diagonal Matrices

ex. $$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -0.4 & 0 & 0 \\ 0 & 0 & 22 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Diagonal Matrices

ex. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -0.4 & 0 & 0 \\ 0 & 0 & 22 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Def. A matrix $A \in \mathbb{R}^{n \times n}$ is **diagonal** if

$i \neq j$ if and only if $A_{ij} = 0$

Diagonal Matrices

ex. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -0.4 & 0 & 0 \\ 0 & 0 & 22 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Def. A matrix $A \in \mathbb{R}^{n \times n}$ is **diagonal** if

$i \neq j$ if and only if $A_{ij} = 0$

Only the diagonal entries can be nonzero

Diagonal Matrices

ex. $$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -0.4 & 0 & 0 \\ 0 & 0 & 22 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Def. A matrix $A \in \mathbb{R}^{n \times n}$ is **diagonal** if

$$i \neq j \quad \text{if and only if } A_{ij} = 0$$

Only the diagonal entries can be nonzero

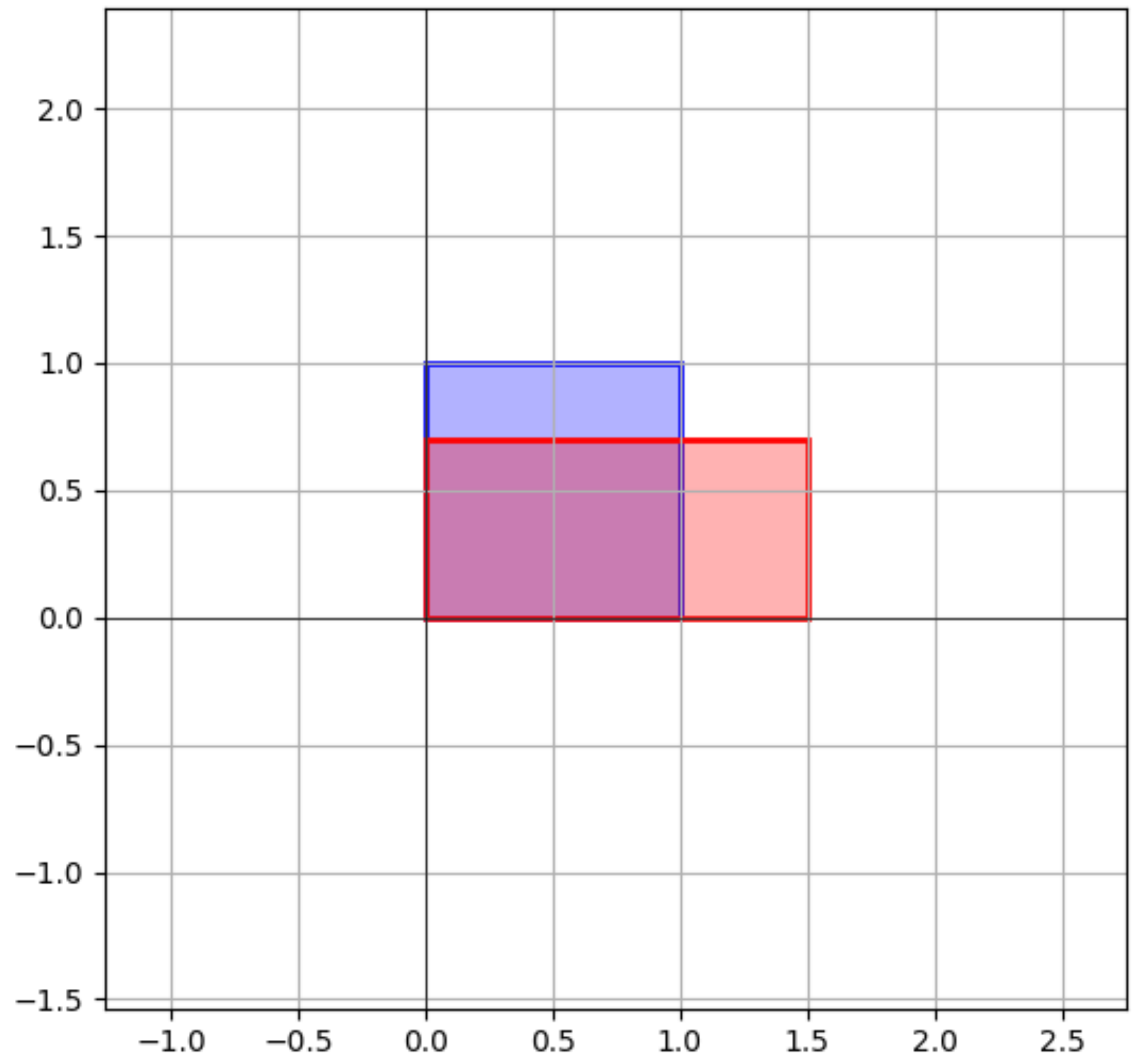
Diagonal matrices are scaling matrices

Recall: Unequal Scaling

The scaling matrix *affects each component of a vector in a simple way*

The diagonal entries <u>scale</u> each corresponding entry

$$\begin{bmatrix} 1.5 & 0 \\ 0 & 0.7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1.5x \\ 0.7y \end{bmatrix}$$



$$\begin{bmatrix} 1.5 & 0 \\ 0 & 0.7 \end{bmatrix}$$

When do matrices "behave" like scaling matrices "up to" change of basis?

Scaling and Eigenvectors

Scaling and Eigenvectors

The idea. Matrices behave like scaling matrices on eigenvectors

$$\begin{bmatrix} c & 0 \\ 0 & d \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c & 0 \\ 0 & d \end{bmatrix}(x\vec{e}_1 + y\vec{e}_2)$$

$$= xc\vec{e}_1 + yd\vec{e}_2$$

$$A(x\underline{\vec{b}_1} + y\underline{\vec{b}_2}) = xA\vec{b}_2 + yA\vec{b}_2$$

eigenvectors

$$= x\lambda_1\vec{b}_2 + x\lambda_2\vec{b}_2$$

Can we expose this behavior in terms of a matrix factorization?

Recall: Matrix Factorization

A **factorization** of a matrix A is an equation which expresses A as a product of one or more matrices, e.g.,

$$A = PBP^{-1}$$

Similar Matrices

$A \sim B$

$\Downarrow$

$B \sim A$

$(P^{-1}) A (P^{-1})^{-1}$

$A = PBP^{-1}$

$\Rightarrow P^{-1}AP = B$

$$A = PB\boxed{P^{-1}}$$

change of basis

Def. A matrix A is **similar** to a matrix B if there is some invertible matrix P such that $A = PBP^{-1}$

A and B are the same up to a change of basis

Similar Matrices and Eigenvalues

Thm. Similar matrices have the *same eigenvalues*

$\text{Nul}(A - \lambda I) \ni \vec{r} \neq \vec{0}$

$B = PAP^{-1}$

$$\det(B - \lambda I) = \det(PAP^{-1} - \lambda I)$$
$$= \det(PAP^{-1} - P\lambda I P^{-1})$$
$$= \det(P(A - \lambda I)P^{-1})$$
$$= \cancel{\det(P)}\,\det(A - \lambda I)\,\cancel{\det(P^{-1})}$$

Diagonalizable Matrices

Diagonalizable Matrices

Def. A matrix A is **diagonalizable** if it is similar to a diagonal matrix

Diagonalizable Matrices

Def. A matrix A is **diagonalizable** if it is similar to a diagonal matrix

There is an invertible matrix P and diagonal matrix D such that $A = PDP^{-1}$

Diagonalizable Matrices

Def. A matrix A is **diagonalizable** if it is similar to a diagonal matrix

There is an invertible matrix P *and diagonal matrix* D *such that* $A = PDP^{-1}$

Diagonalizable matrices are the same as scaling matrices up to a change of basis

Important: Not all Matrices are Diagonalizable

This is very different from the LU factorization

We will need to figure out which matrices are diagonalizable

Application: Matrix Powers

only take the power of B

Thm. If $A = PBP^{-1}$, then $A^k = PB^kP^{-1}$

It may be easier to take the power of B (as in the case of diagonal matrices)

$$A^2 = PB\cancel{P^{-1}}\cancel{P}BP^{-1}$$

$$= PB^2P^{-1}$$

But how do we find the diagonalization...

Diagonalization and Eigenvectors

Suppose we have a diagonalization

$$A = PDP^{-1}$$

What do we know about it?

Columns of P are eigenvectors

$$A = [\mathbf{p}_1\ \mathbf{p}_2\ \mathbf{p}_3]\begin{bmatrix}\lambda_1 & 0 & 0\\ 0 & \lambda_2 & 0\\ 0 & 0 & \lambda_3\end{bmatrix}[\mathbf{p}_1\ \mathbf{p}_2\ \mathbf{p}_3]^{-1}$$

$$PDP^{-1}\vec{p}_1 = PD\vec{e}_1 = P\lambda_1\vec{e}_1$$

$$= \lambda_1\vec{p}_1$$

Columns of P are eigenvectors

$$A = [\mathbf{p}_1 \ \mathbf{p}_2 \ \mathbf{p}_3] \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} [\mathbf{p}_1 \ \mathbf{p}_2 \ \mathbf{p}_3]^{-1}$$

Columns of P are eigenvectors

$$A = [\mathbf{p}_1\ \mathbf{p}_2\ \mathbf{p}_3]\begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}[\mathbf{p}_1\ \mathbf{p}_2\ \mathbf{p}_3]^{-1}$$

In fact, the columns of P form an **eigenbasis** of $\mathbb{R}^n$ for A

Columns of P are eigenvectors

$$A = [\mathbf{p}_1 \ \mathbf{p}_2 \ \mathbf{p}_3] \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} [\mathbf{p}_1 \ \mathbf{p}_2 \ \mathbf{p}_3]^{-1}$$

In fact, the columns of P form an **eigenbasis** of $\mathbb{R}^n$ for A

And the entries of D are the **eigenvalues** associated to each eigenvector

Columns of P are eigenvectors

$$A = \overset{\text{eigenbasis}}{[\mathbf{p}_1\ \mathbf{p}_2\ \mathbf{p}_3]} \overset{\text{eigenvalues}}{\begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}} [\mathbf{p}_1\ \mathbf{p}_2\ \mathbf{p}_3]^{-1}$$

In fact, the columns of P form an **eigenbasis** of $\mathbb{R}^n$ for A

And the entries of D are the **eigenvalues** associated to each eigenvector

A diagonalization exposes a lot of information about A

The Diagonalization Theorem

The Diagonalization Theorem

Thm. A matrix is diagonalizable if and only if it has an eigenbasis

The Diagonalization Theorem

Thm. A matrix is diagonalizable if and only if it has an eigenbasis

We can use the same recipe to go in the other direction, given an eigenbasis, we can **build a diagonalization**

Diagonalizing a Matrix

High Level

$$A = PDP^{-1}$$

High Level

$$A = PDP^{-1}$$

The columns of P form an *eigenbasis* for A

High Level

$$A = PDP^{-1}$$

The columns of P form an *eigenbasis* for A

The diagonal of D are the eigenvalues for each column of P

High Level

$$A = PDP^{-1}$$

The columns of P form an *eigenbasis* for A

The diagonal of D are the eigenvalues for each column of P

The matrix P^{-1} is a change of basis to this eigenbasis of A

Step 1: Eigenvalues

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & 3 \\ 3 & 3 & 1 \end{bmatrix}$$

Find all the eigenvalues of A

Find the roots of $\det(A - \lambda I)$

e.g. $\det(A - \lambda I) = -(\lambda - 1)(\lambda + 2)^2$

Step 2: Eigenvectors

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & 3 \\ 3 & 3 & 1 \end{bmatrix}$$

$$\lambda_1 = 1$$

$$\lambda_2 = -2$$

Find **bases** of the corresponding eigenspaces

e.g.

$$\mathrm{Nul}(A - I) = \mathrm{span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

$$\mathrm{Nul}(A + 2I) = \mathrm{span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Step 3: Construct P

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & 3 \\ 3 & 3 & 1 \end{bmatrix}$$

$$\lambda_1 = 1$$

$$\lambda_2 = -2$$

$$\text{Nul}(A - I) = \text{span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

$$\text{Nul}(A + 2I) = \text{span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

If there are n eigenvectors from the previous step they form an **eigenbasis**

Build the matrix with these vectors as the columns

e.g.

$$P = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Step 5: Construct D

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & 3 \\ 3 & 3 & 1 \end{bmatrix}$$

$$\lambda_1 = 1$$

$$\lambda_2 = -2$$

$$P = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Build the matrix with eigenvalues as diagonal entries

Note the order. It should be the same as the order of columns of P

e.g. 1 2-2 3-2

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

Step 6: Invert P

Find the inverse of P

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & 3 \\ 3 & 3 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Putting it Together

$$\underset{A}{\begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & 3 \\ 3 & 3 & 1 \end{bmatrix}} = \underset{P}{\begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}} \underset{D}{\begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}} \underset{P^{-1}}{\begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}^{-1}}$$

We know how to do every step, its a matter of putting it all together

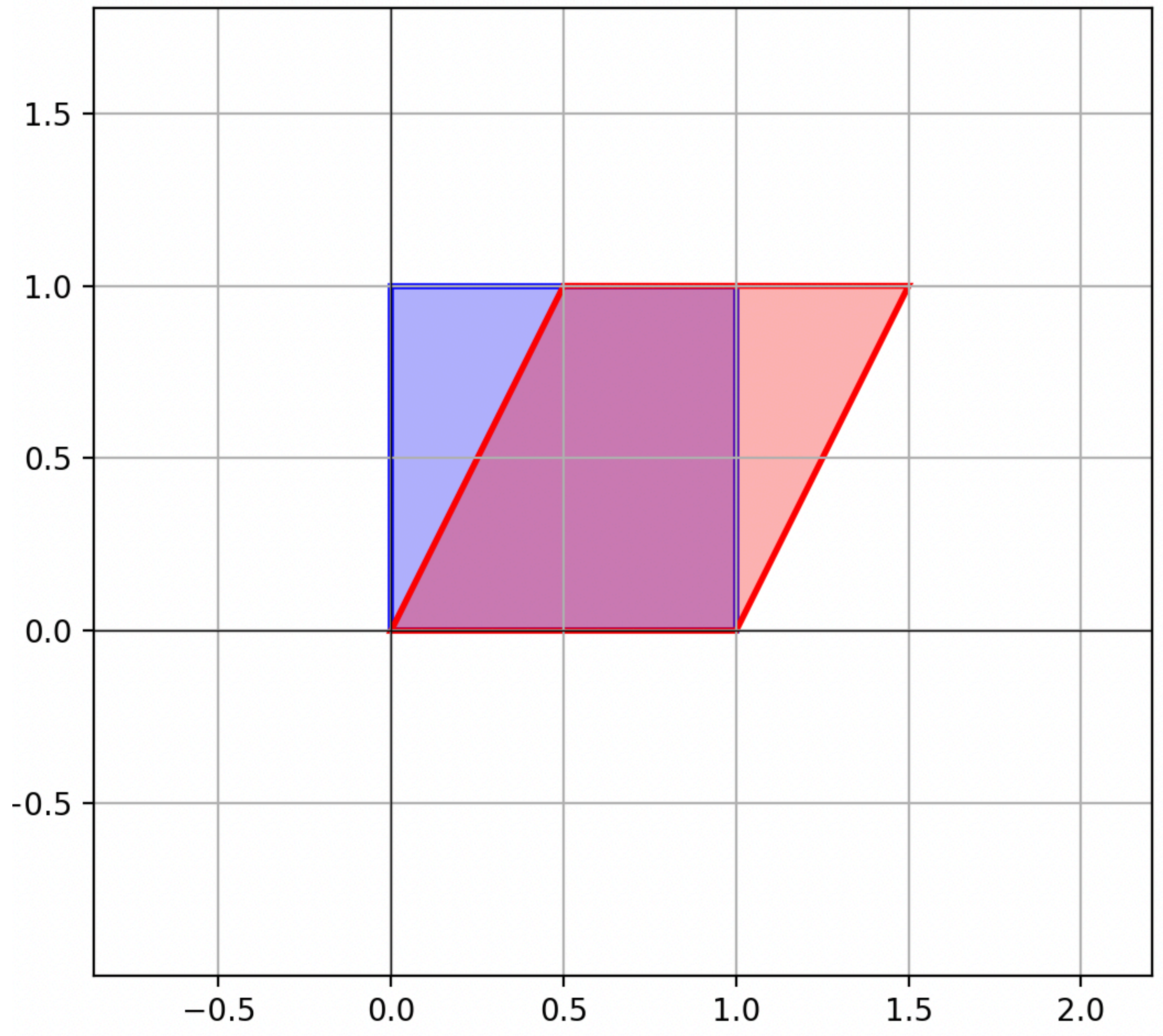
Example of Failure: Shearing

$$A = \begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix}$$

The shearing matrix has a single eigenvalue with an eigenspace of dimension 1

We can't build an eigenbasis of $\mathbb{R}^2$ for A

In other words, A is not diagonalizable



Important case: Distinct Eigenvalues

$$\begin{bmatrix} 1 & -3 & 4 & 2 \\ 0 & -2 & 3 & -1 \\ 0 & 0 & 10 & 5 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

Thm. If a matrix in $\mathbb{R}^{n\times n}$ has n distinct eigenvalues, then it is diagonalizable

This is because eigenvectors with distinct eigenvalues are *linearly independent*

Example

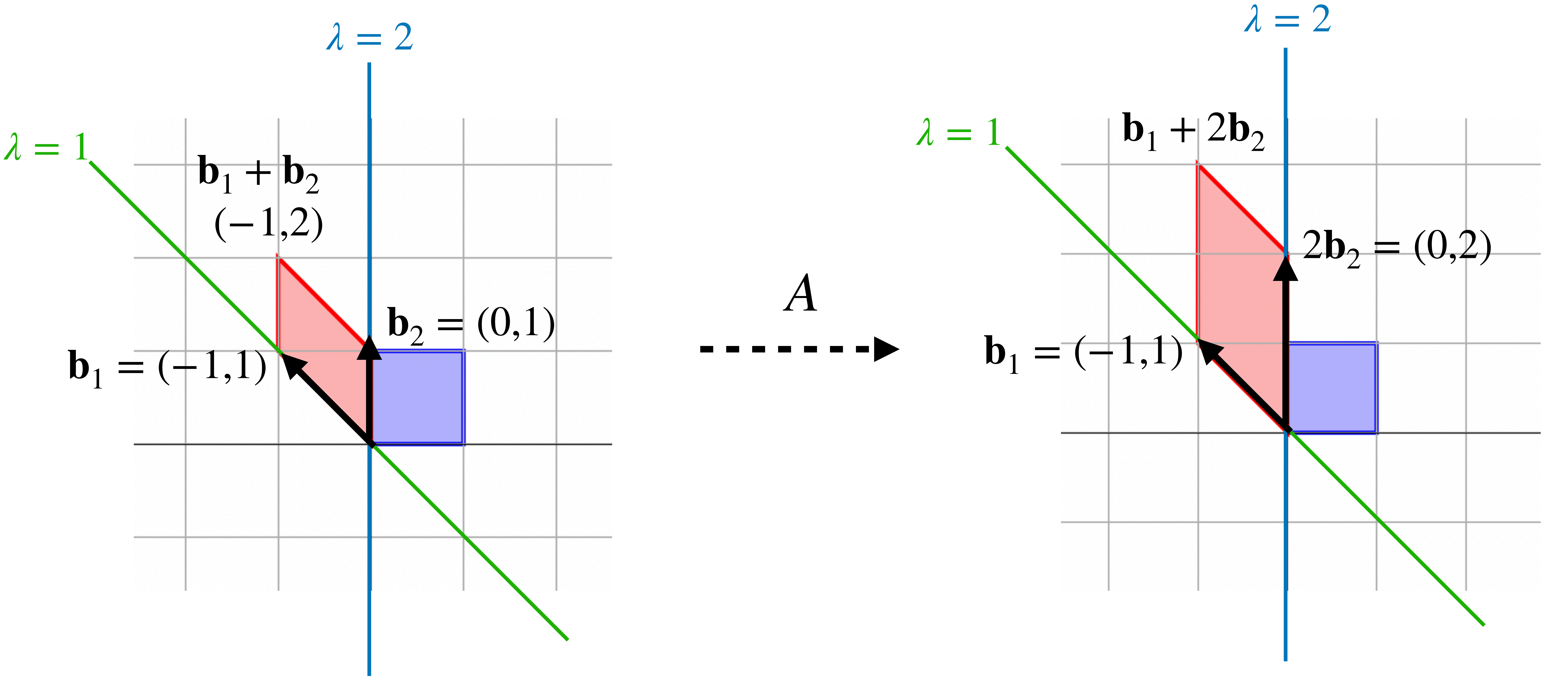
$$\begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}$$

Find a diagonalization of the above matrix

The Picture

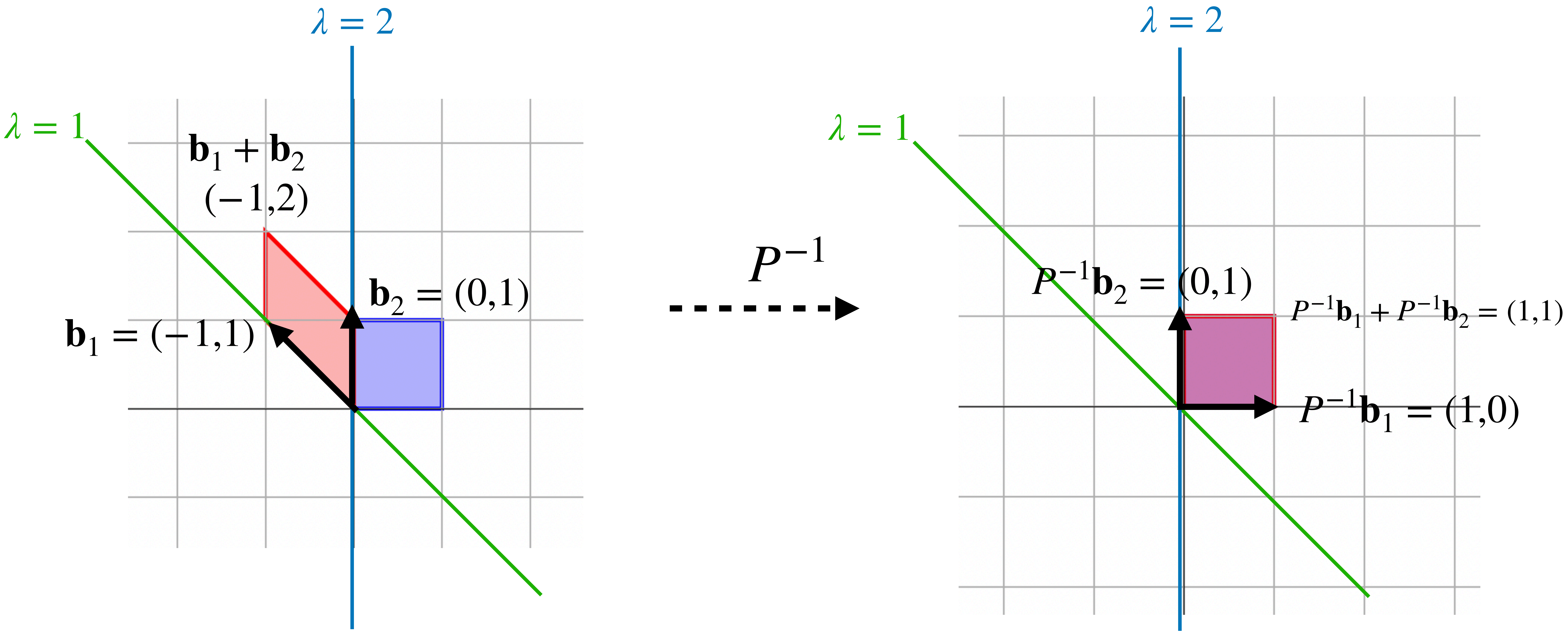
Example (Geometric)

$$A = \begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix}$$



Example (Geometric)

$$P^{-1} = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}^{-1}$$

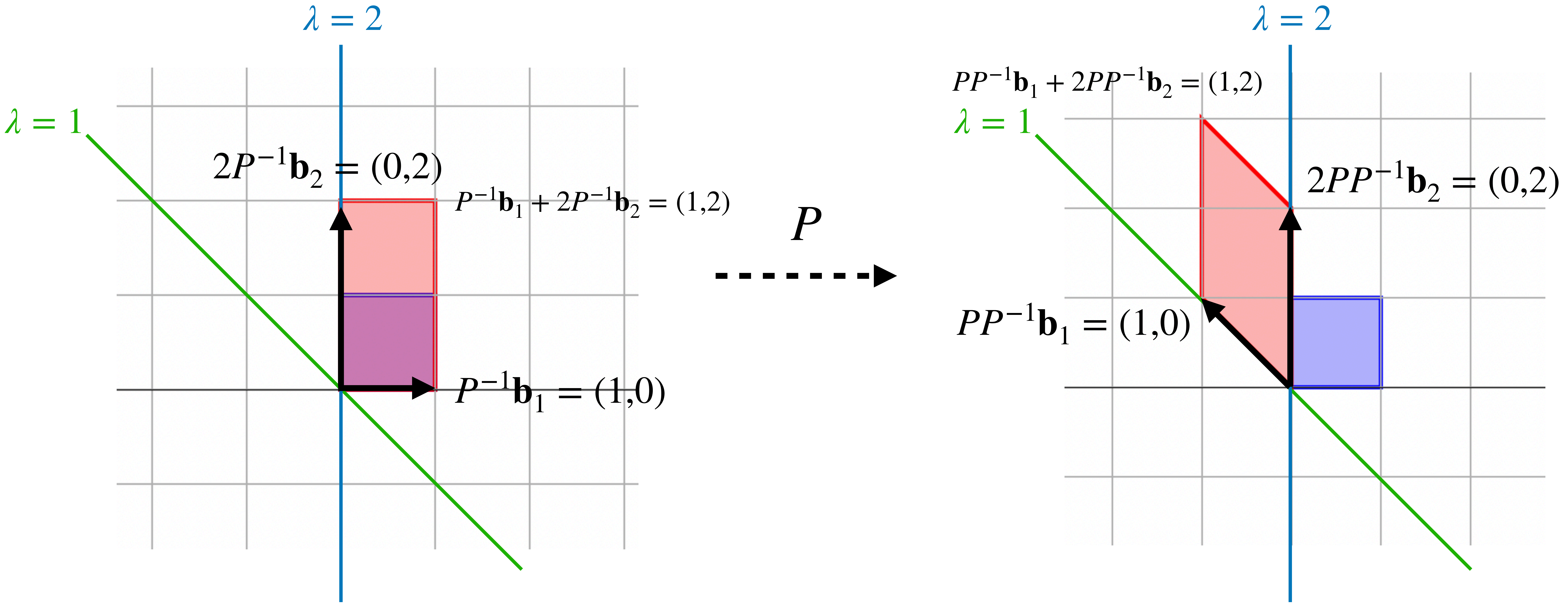


Example (Geometric)

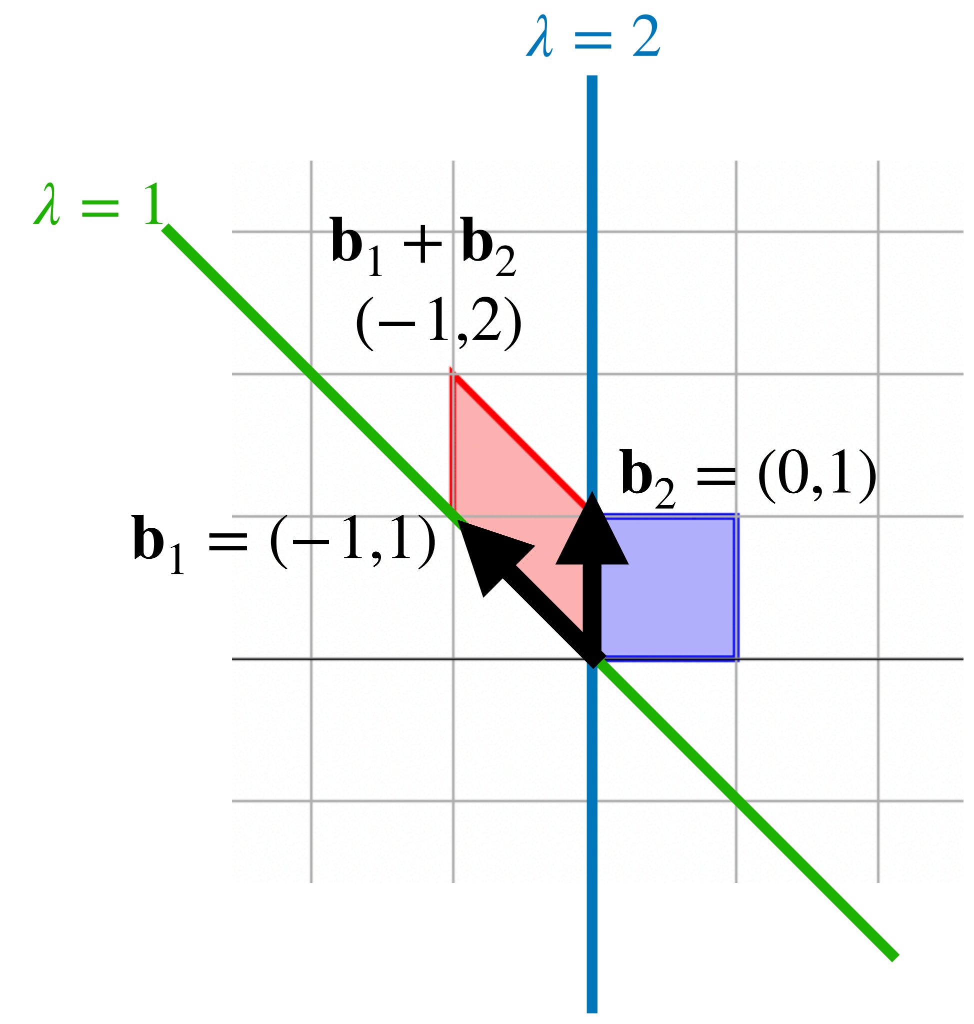
$$D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

Example (Geometric)

$$P = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}$$



Example (Geometric)



$$A = PDP^{-1}$$

$$\begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}^{-1}$$